

AquaMorph: Emulating Marine Animals with Soft Robotics

Capstone Senior Design Project • Spring 2025



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Diving into the Problem



Oceans cover over 70% of Earth's surface, yet much of it remains unexplored. The tools we rely on are outdated: rigid, loud, and ill-equipped to move like the marine life they're meant to study.

Traditional robots fall short — ***we need systems that move with the water, not against it.***

The Problem

Rigid frames
Noisy propellers
Poor maneuverability
Harms marine life

The Need

Quiet, flexible movement
Biomimetic propulsion
Safe for marine ecosystem
Compatible with divers and sensors

Our Solution

Fin based, lifelike swimming
Lightweight and field deployable
Built for low impact observation

Reeling in the Requirements

- **Biomimetic Motion:** ≥ 0.25 m/s, neutrally buoyant at 10m.
- **Low Disturbance:** < 70 dB, 80% visual similarity to marine life
- **Underwater Ready:** Waterproof to 20m, 60 min runtime
- **User Friendly:** ≤ 5 kg, ≤ 2 operators, intuitive controls
- **Data Driven:** Photo/video, obstacle detection within 5m, onboard sensors
- **Built to Last:** Lifespan of ≥ 1.5 years with durable and portable design

The Other Fish in the Sea

The field of bioinspired aquatic robotics is still emerging, with limited direct competition currently in the market.

However, a few comparable technologies demonstrate growing interest and potential in this space:

In the Market	In Research
iBubble  • Autonomous underwater drone designed for divers • Rigid structure limits biomimetic movement and marine life interaction	SoFi (MIT CSAIL)  • Soft robotic fish for marine observation • Biomimetic swimming and flexible actuation • Not available for consumer or field deployment • Limited maneuverability and short battery life

Sailing to a Solution



Manta Inspired Movement :

- Pulley-driven fins and tail mimic real marine life movement

Soft Actuators:

- Silicone body and servo motors create smooth, low-noise movement.

Built for the deep:

- Waterproof electronics, onboard sensors, and a low-light camera.

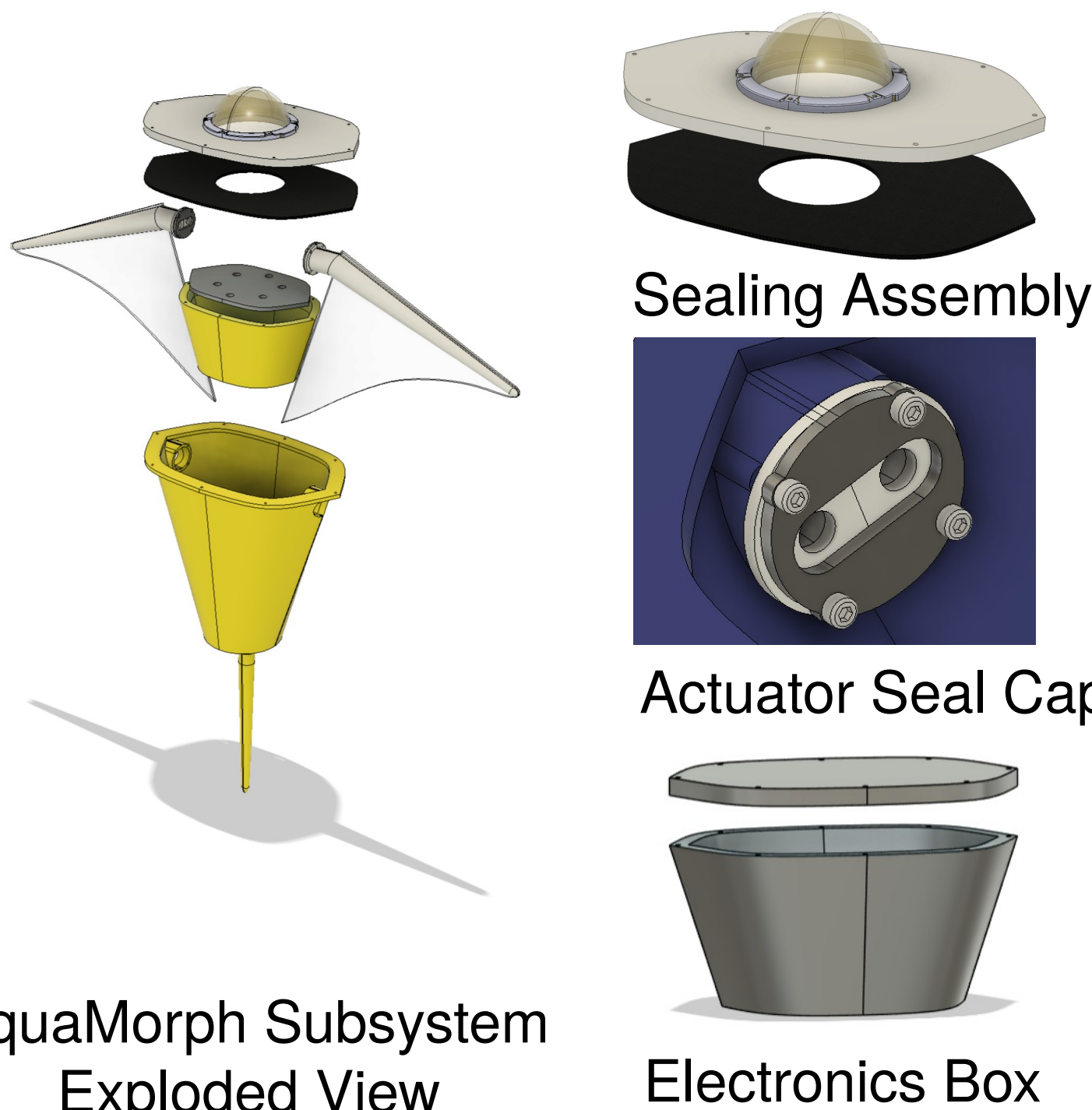
Lightweight and Intuitive:

- Easy to carry and driver operated

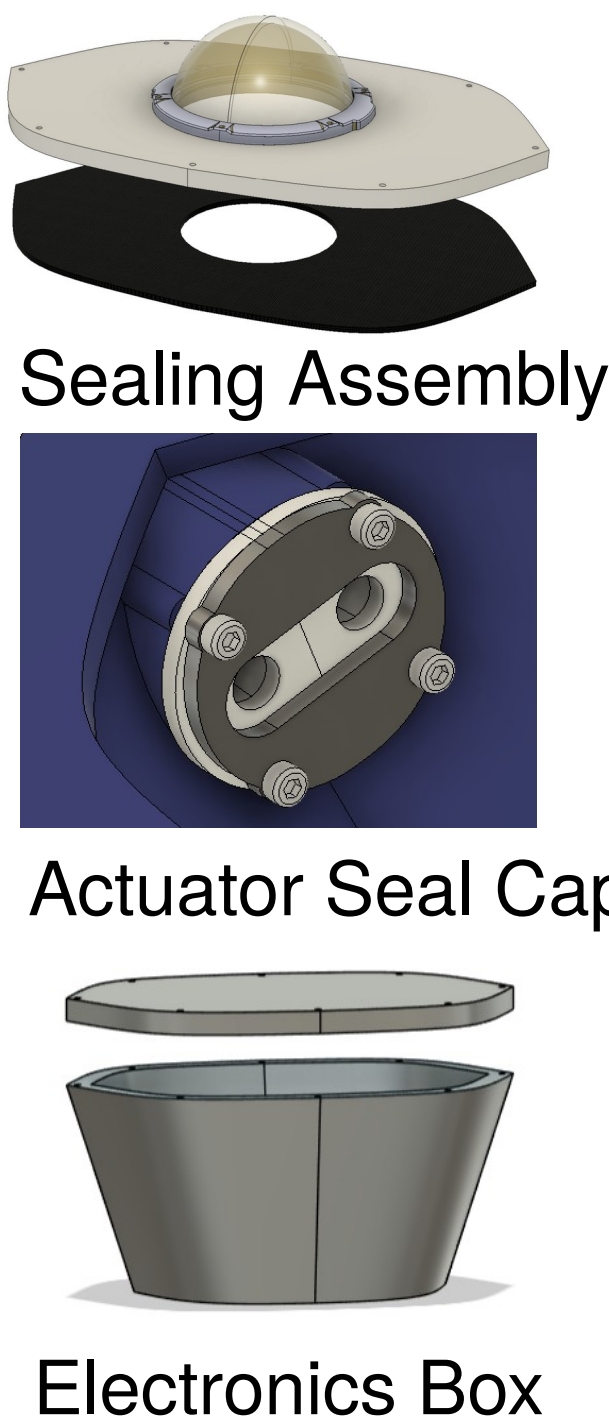
Designing for the Deep

System Features Breakdown:

- Fully enclosed, gasketed body, waterproof electronics box.
- Camera and sensor package for data collection.
- App compatibility with Bluetooth connection.
- Servo-driven soft robotic actuators for wing and tail movement for locomotion.
- Waterproof epoxy coating for body.



AquaMorph Subsystem Exploded View



Sealing Assembly

Actuator Seal Cap

Electronics Box

How we Make a Splash

Our definition of innovation is making waves where no one else is looking.

We accomplished this by combining soft robotics and marine biology to create a manta-inspired robot that moves like nature, not a machine — unlocking new possibilities in underwater research and diver assistance

What makes us different:



Biomimetic Swimming: Fin-based propulsion mimics manta ray locomotion.



Soft & Silent: Silicone actuators reduce noise and impact on marine life.



Stable Data Capture: Smooth motion enhances sensor accuracy.



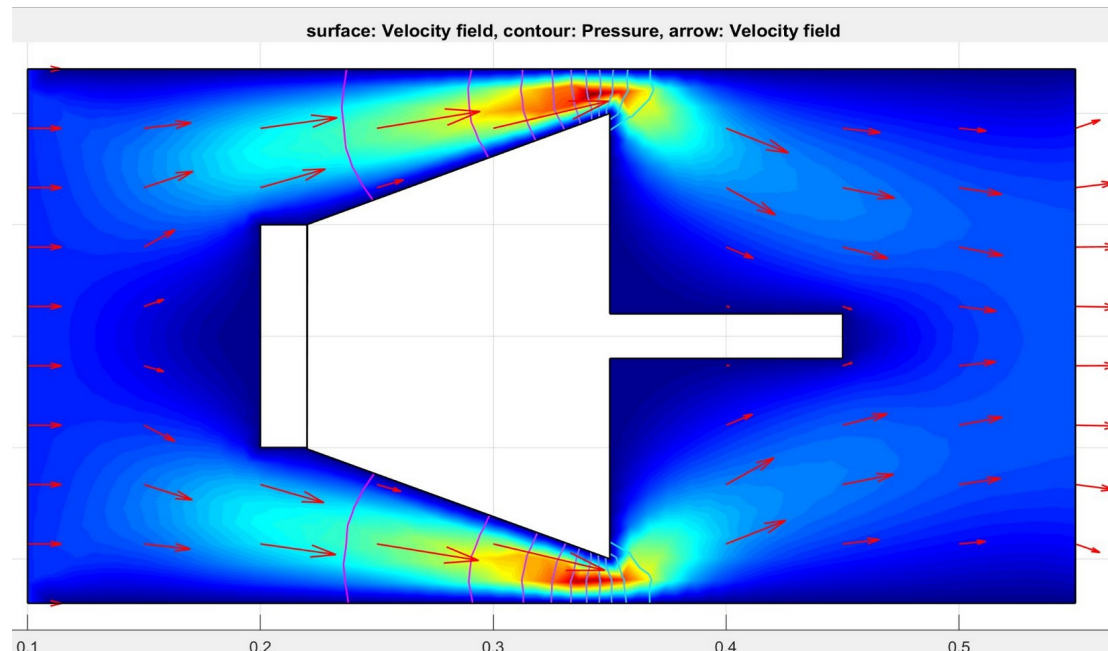
Diver-Oriented Design: Built to assist in low-visibility environments.



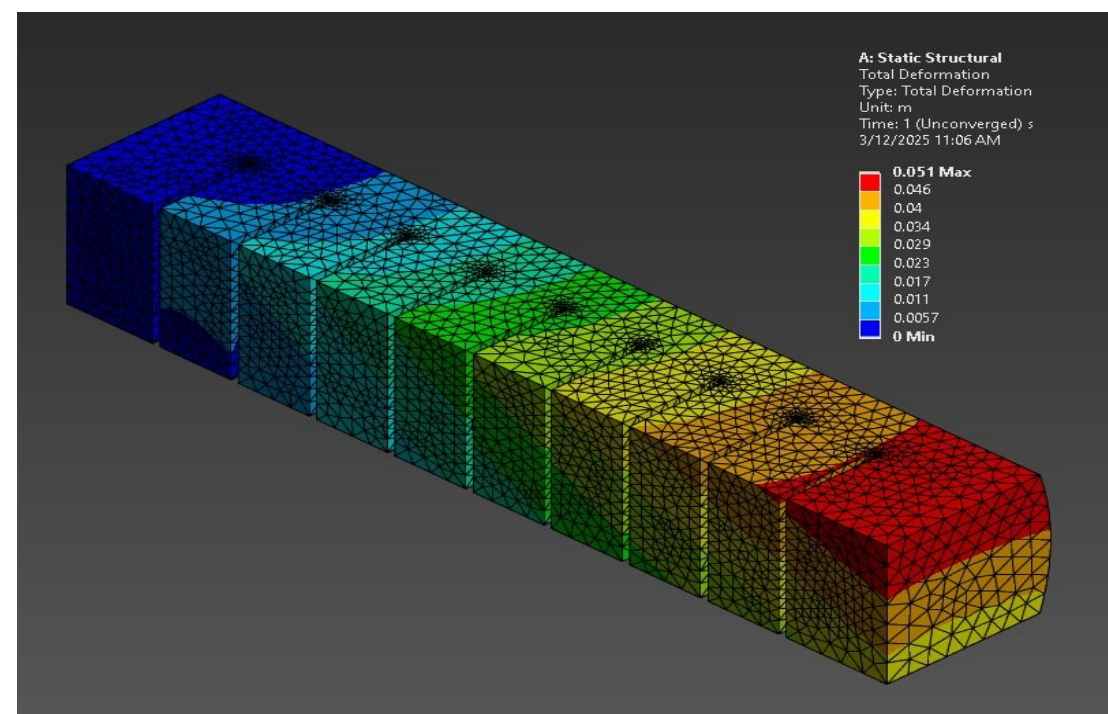
Modular & Lightweight: Ready for rapid iteration and field use.

Testing the Waters

FEA & CFD

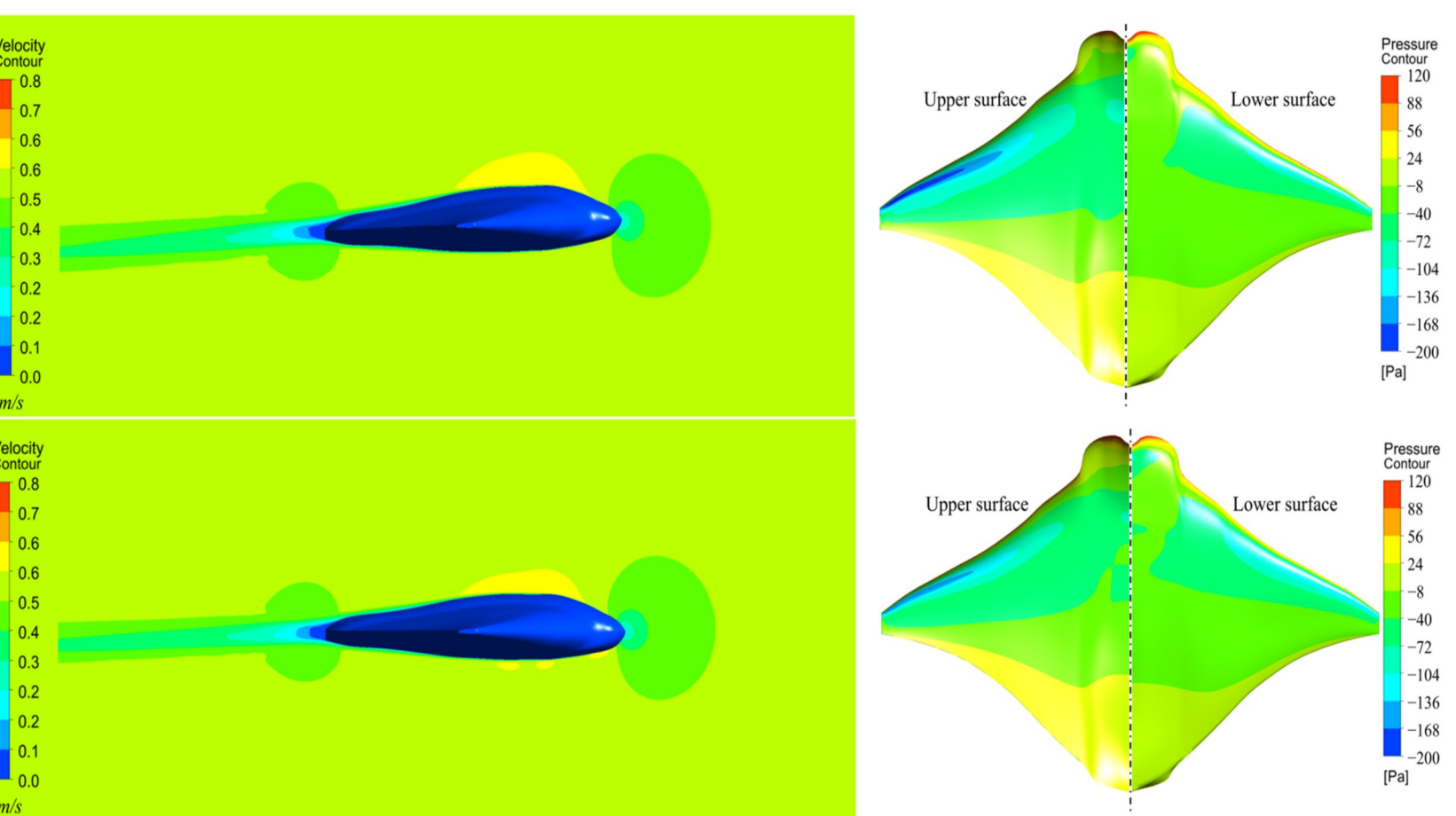


Simplified Top-Down Velocity & Pressure Field



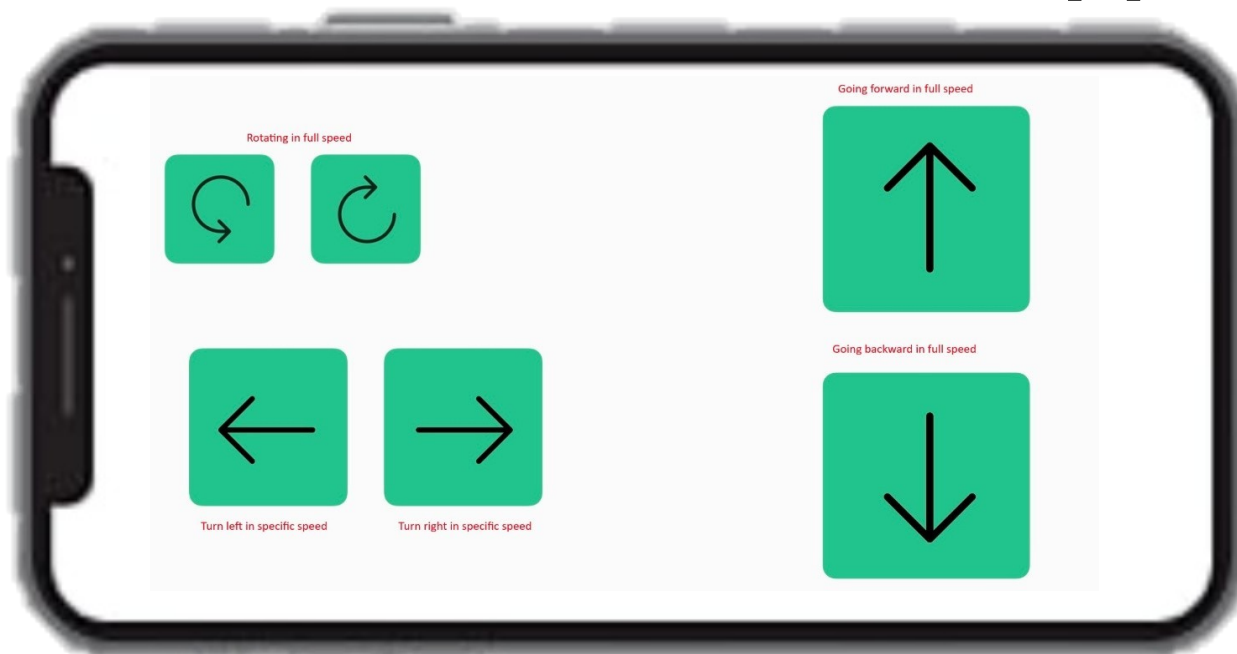
Static & Total Deformation on Actuators

Underwater Testing

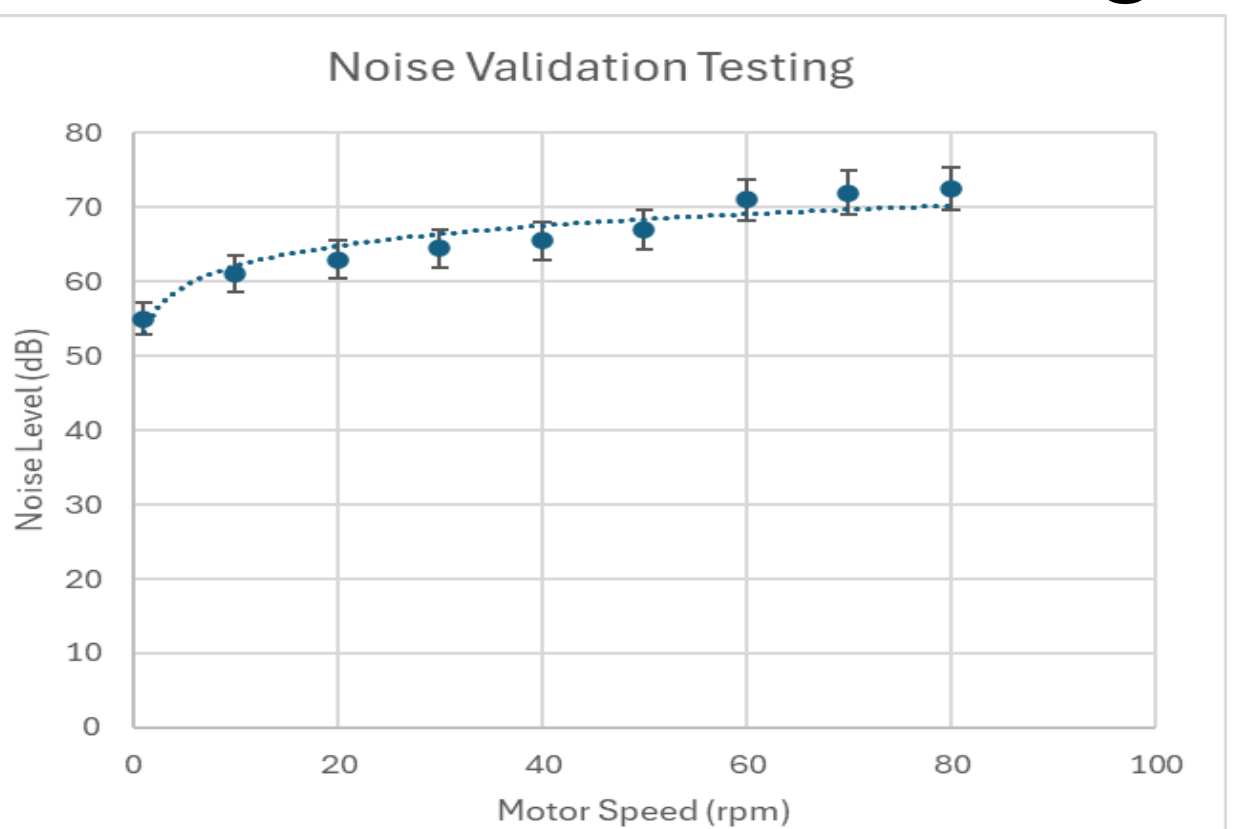


Manta Ray Hydrodynamics
Ma, Y.; Huang, Investigation of the Hydrodynamic Characteristics of Two Manta Rays Tandem Gliding. *J. Mar. Sci. Eng.* 2022

Remote Control via App



Noise Validation Testing



Swimming Towards the Future



- **Buoyancy Control:** Implement component to adjust buoyancy and control depth of robot.
- **Increasing Longevity:** Use higher quality materials to increase durability and improve waterproofing at higher depths for longer times.

- **Increasing Range:** Implement acoustic modem for long range underwater communication.
- **Improve Design:** Refine body and actuator designs to improve hydrodynamics and appearance.
- **Increasing Capabilities:** Integrating more advanced control systems and additional sensors for more precise movement.
- **Real-World Testing:** Field test in natural aquatic environments to assess performance and environmental adaptability.